

- * Thus, extended-release valproate is dosed approximately 8–20% higher when converting patients to the ER formulation
- Depakote (divalproex sodium) is an enteric-coated stable compound containing both valproic acid and sodium valproate
- * Divalproex immediate-release formulation reduces gastrointestinal side effects compared to generic valproate
- * Divalproex ER improves gastrointestinal side effects and alopecia compared to immediate-release divalproex or generic valproate
- The amide of valproic acid is available in Europe [valpromide (Depamide)]
- Trough plasma drug levels $>45 \mu\text{g/mL}$ may be required for either antimanic effects or anticonvulsant actions
- Trough plasma drug levels up to $100 \mu\text{g/mL}$ are generally well tolerated
- Trough plasma drug levels up to $125 \mu\text{g/mL}$ may be required in some acutely manic patients
- Dosages to achieve therapeutic plasma levels vary widely, often between 750–3000 mg/day

Overdose

- Fatalities have been reported; coma, restlessness, hallucinations, sedation, heart block

Long-Term Use

- Requires regular liver function tests and platelet counts

Habit Forming

- No

How to Stop

- Taper; may need to adjust dosage of concurrent medications as valproate is being discontinued
- Patients may seize upon withdrawal, especially if withdrawal is abrupt
- * Rapid discontinuation increases the risk of relapse in bipolar disorder
- Discontinuation symptoms uncommon

Pharmacokinetics

- Mean terminal half-life 9–16 hours
- Metabolized primarily by the liver, approximately 25% dependent upon CYP450 system (CYP450 2C9 and 2C19)
- Inhibits CYP450 2C9
- Food slows rate but not extent of absorption



Drug Interactions

- * Lamotrigine dose should be reduced by perhaps 50% if used with valproate, as valproate inhibits metabolism of lamotrigine and raises lamotrigine plasma levels, theoretically increasing the risk of rash
- Plasma levels of valproate may be lowered by carbamazepine, phenytoin, ethosuximide, phenobarbital, rifampin
 - Aspirin may inhibit metabolism of valproate and increase valproate plasma levels
 - Plasma levels of valproate may also be increased by felbamate, chlorpromazine, fluoxetine, fluvoxamine, topiramate, cimetidine, erythromycin, and ibuprofen
 - Valproate inhibits metabolism of ethosuximide, phenobarbital, and phenytoin, and can thus increase their plasma levels
 - No likely pharmacokinetic interactions of valproate with lithium or atypical antipsychotics
 - Use of valproate with clonazepam may cause absence status
 - Reports of hypothermia and hyperammonemia with or without encephalopathy in patients taking topiramate combined with valproate, though this is not due to a pharmacokinetic interaction; in patients who develop unexplained lethargy, vomiting, or change in mental status, an ammonia level should be measured



Other Warnings/Precautions

* Be alert to the following symptoms of hepatotoxicity that require immediate attention: malaise, weakness, lethargy, facial edema,

anorexia, vomiting, yellowing of the skin and eyes

* Be alert to the following symptoms of pancreatitis that require immediate attention: abdominal pain, nausea, vomiting, anorexia

* Teratogenic effects in developing fetuses such as neural tube defects may occur with valproate use

* Somnolence may be more common in the elderly and may be associated with dehydration, reduced nutritional intake, and weight loss, requiring slower dosage increases, lower doses, and monitoring of fluid and nutritional intake

- Use in patients with thrombocytopenia is not recommended; patients should report easy bruising or bleeding
- Evaluate for urea cycle disorders, as hyperammonemic encephalopathy, sometimes fatal, has been associated with valproate administration in these uncommon disorders; urea cycle disorders, such as ornithine transcarbamylase deficiency, are associated with unexplained encephalopathy, mental retardation, elevated plasma ammonia, cyclical vomiting, and lethargy
- Valproate is associated with a rare but serious skin condition known as drug reaction with eosinophilia and systemic symptoms (DRESS). DRESS may begin as a rash but can progress to other parts of the body and can include symptoms such as fever, swollen lymph nodes, inflammation of organs, and an increase in white blood cells known as eosinophilia. In some cases, DRESS can lead to death

- Whenever possible, warn patients and their caregivers about the possibility of activating side effects, and advise them to report such side effects immediately

Do Not Use

- If patient has pancreatitis
- If patient has serious liver disease
- If patient has urea cycle disorder
- If there is a proven allergy to valproic acid, valproate, or divalproex

Special Populations

Renal Impairment

- No dose adjustment necessary

Hepatic Impairment

- Contraindicated

Cardiac Impairment

- No dose adjustment necessary

Elderly

- Reduce starting dose and titrate slowly; dosing is generally lower than in healthy adults

- * Sedation in the elderly may be more common and associated with dehydration, reduced nutritional intake, and weight loss
- Monitor fluid and nutritional intake
- 1 in 3 elderly patients in long-term care who receive valproate may ultimately develop thrombocytopenia

Children and Adolescents

- * Not generally recommended for use in children under age 10 for bipolar disorder except by experts and when other options have been considered
- Children under age 2 have significantly increased risk of hepatotoxicity, as they have a markedly decreased ability to eliminate valproate compared to older children and adults
- Use requires close medical supervision

Pregnancy

- Effective June 30, 2015, the US FDA requires changes to the content and format of pregnancy and lactation information in prescription drug labels, including the elimination of the pregnancy letter categories; the Pregnancy and Lactation Labeling Rule (PLLR or final rule) applies only to prescription drugs and will be phased in gradually for drugs approved on or after June 30, 2001
- * Use during first trimester may raise risk of neural tube defects (e.g., spina bifida) or other congenital anomalies

- Cases of developmental delay in the absence of teratogenicity associated with fetal exposure have been identified
- Increased risk of lower cognitive test scores in children whose mothers took valproate during pregnancy
- Use in women of childbearing potential requires weighing potential benefits to the mother against the risks to the fetus
- * If drug is continued, monitor clotting parameters and perform tests to detect birth defects
- * If drug is continued, start on folate 1 mg/day early in pregnancy to reduce risk of neural tube defects
- * If drug is continued, consider vitamin K during the last 6 weeks of pregnancy to reduce risks of bleeding
- Antiepileptic Drug Pregnancy Registry: 1-888-233-2334, www.aedpregnancyregistry.org
- Taper drug if discontinuing
- Seizures, even mild seizures, may cause harm to the embryo/fetus
- * For bipolar patients, valproate should generally be discontinued before anticipated pregnancies
- Recurrent bipolar illness during pregnancy can be quite disruptive
- * For bipolar patients, given the risk of relapse in the postpartum period, mood stabilizer treatment such as valproate should generally be restarted immediately after delivery if patient is unmedicated during pregnancy

* Atypical antipsychotics may be preferable to lithium or anticonvulsants such as valproate if treatment of bipolar disorder is required during pregnancy

- Bipolar symptoms may recur or worsen during pregnancy and some form of treatment may be necessary

Breast Feeding

- Some drug is found in mother's breast milk
- * Generally considered safe to breast feed while taking valproate
- If drug is continued while breast feeding, infant should be monitored for possible adverse effects
- If infant shows signs of irritability or sedation, drug may need to be discontinued
- * Bipolar disorder may recur during the postpartum period, particularly if there is a history of prior postpartum episodes of either depression or psychosis
- * Relapse rates may be lower in women who receive prophylactic treatment for postpartum episodes of bipolar disorder
- Atypical antipsychotics and anticonvulsants such as valproate may be safer than lithium during the postpartum period when breast feeding

The Art Of Psychopharmacology

Potential Advantages

- Manic phase of bipolar disorder
- Works well in combination with lithium and/or atypical antipsychotics
- Patients for whom therapeutic drug monitoring is desirable

Potential Disadvantages

- Depressed phase of bipolar disorder
- Patients unable to tolerate sedation or weight gain
- Multiple drug interactions
- Multiple side effect risks
- Pregnant patients

Primary Target Symptoms

- Unstable mood
- Incidence of migraine
- Incidence of partial complex seizures



Pearls (for bipolar disorder)

* Valproate is a first-line treatment option that may be best for patients with mixed states of bipolar disorder or for patients with rapid-cycling bipolar disorder

* Seems to be more effective in treating manic episodes than depressive episodes in bipolar disorder (treats from above better than it treats from below)

* May also be more effective in preventing manic relapses than in preventing depressive episodes (stabilizes from above better than it stabilizes from below)

- Only a third of bipolar patients experience adequate relief with a monotherapy, so most patients need multiple medications for best control

- Useful in combination with atypical antipsychotics and/or lithium for acute mania

* May also be useful for bipolar disorder in combination with lamotrigine, but must reduce lamotrigine dose by half when combined with valproate

- Usefulness for bipolar disorder in combination with anticonvulsants other than lamotrigine is not well demonstrated; such combinations can be expensive and are possibly ineffective or even irrational

* May be useful as an adjunct to atypical antipsychotics for rapid onset of action in schizophrenia

* Used to treat aggression, agitation, and impulsivity not only in bipolar disorder and schizophrenia but also in many other disorders, including dementia, personality disorders, and brain injury

- Patients with acute mania tend to tolerate side effects better than patients with hypomania or depression

- Multivitamins fortified with zinc and selenium may help reduce alopecia
- Association of valproate with polycystic ovaries is controversial and may be related to weight gain, obesity, or epilepsy
- Nevertheless, may wish to be cautious in administering valproate to women of childbearing potential, especially adolescent female bipolar patients, and carefully monitor weight, endocrine status, and ovarian size and function

* In women of childbearing potential who are or are likely to become sexually active, should inform about risk of harm to the fetus and monitor contraceptive status

- Association of valproate with decreased bone mass is controversial and may be related to activity levels, exposure to sunlight, and epilepsy, and might be prevented by supplemental vitamin D 2000 IU/day and calcium 600–1000 mg/day
- New delayed-release capsule of valproic acid (Stavzor) may be easier to swallow than other formulations
- A prodrug of valproic acid, valpromide, is available in several European countries
- Although valpromide is rapidly transformed to valproic acid, it has some unique characteristics that can affect drug interactions
- In particular, valpromide is a potent inhibitor of liver microsomal epoxide hydrolase and thus causes clinically significant increases in

the plasma levels of carbamazepine-10,11-epoxide (the active metabolite of carbamazepine)

Suggested Reading

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Varenicline

Therapeutics

Brands

- Chantix
- Champix
- *see index for additional brand names*

Generic?

No



Class

- Neuroscience-based Nomenclature: acetylcholine receptor partial agonist (ACh-RPA)
- Smoking cessation treatment; alpha 4 beta 2 partial agonist at nicotinic acetylcholine receptors

Commonly Prescribed for

(bold for FDA approved)

- **Nicotine addiction/dependence**



How the Drug Works

- Causes sustained but small amounts of dopamine release (less than with nicotine)
- Specifically, as a partial agonist at alpha 4 beta 2 nicotinic acetylcholine receptors, varenicline activates these receptors to a lesser extent than the full agonist nicotine and also prevents nicotine from binding to these receptors
- Most prominent actions are on mesolimbic dopaminergic neurons in the ventral tegmental area

How Long Until It Works

- Recommended initial treatment trial is 12 weeks; an additional 12-week trial in individuals who stop smoking after 12 weeks may increase likelihood of long-term abstinence

If It Works

- Reduces withdrawal symptoms and the urge to smoke; increases abstinence

If It Doesn't Work

- Evaluate for and address contributing factors, then reattempt treatment
- Consider switching to another agent



Best Augmenting Combos for Partial Response or Treatment Resistance

- Best to attempt other monotherapies

Tests

- None for healthy individuals

Side Effects

How Drug Causes Side Effects

- Theoretically due to increases in dopamine concentrations at receptors in parts of the brain and body other than those that cause therapeutic actions

Notable Side Effects

- Dose-dependent nausea, vomiting, constipation, flatulence
- Insomnia, headache, abnormal dreams



Life-Threatening or Dangerous Side Effects

- Rare activation of agitation, depressed mood, suicidal ideation, suicidal behavior
- Rare seizures

Weight Gain



- Reported but not expected

- Some patients report weight loss

Sedation



- Reported but not expected
- Some patients report activation and insomnia

What to Do About Side Effects

- Wait
- Adjust dose
- If side effects persist, discontinue use

Best Augmenting Agents for Side Effects

- Dose reduction or switching to another agent may be more effective since most side effects cannot be improved with an augmenting agent

Dosing And Use

Usual Dosage Range

- 1 mg twice daily

Dosage Forms

- Tablet 0.5 mg, 1 mg

How to Dose

- Patient should either set a quit date and then begin taking varenicline 1 week before the quit date or start varenicline and then quit smoking between days 8 and 35 of treatment
- Initial 0.5 mg/day; after 3 days increase to 1 mg/day in two divided doses; after 4 days can increase to 2 mg/day in two divided doses



Dosing Tips

- Varenicline should be taken following a meal and with a full glass of water
- Providing educational materials and counseling in combination with varenicline treatment can increase the chances of success
- Initial recommended treatment duration is 12 weeks; for individuals who have stopped smoking after 12 weeks continued treatment for an additional 12 weeks can increase the likelihood of long-term abstinence
- For patients who are unsuccessful in their attempt to quit following 12 weeks of treatment or those who relapse following treatment, it is best to attempt to address factors contributing to the failed attempt and then reintroduce treatment

Overdose

- Limited available data

Long-Term Use

- Treatment for up to 24 weeks has been found effective

Habit Forming

- No

How to Stop

- Taper to avoid withdrawal effects, but no well-documented tolerance, dependence, or withdrawal reactions

Pharmacokinetics

- Elimination half-life 24 hours



Drug Interactions

- Does not inhibit hepatic enzymes or renal transport proteins, and thus is unlikely to affect plasma concentrations of other drugs
- Is not hepatically metabolized and thus is unlikely to be affected by other drugs
- Side effects may be increased if varenicline is taken with nicotine replacement therapy
- Varenicline may alter one's reaction to alcohol, with case reports showing decreased tolerance to alcohol, increased drunkenness, unusual or aggressive behavior, or no memory of things that happened

Other Warnings/Precautions

- Monitor patients for changes in behavior, agitation, depressed mood, worsening of preexisting psychiatric illness, and suicidality
- Use cautiously in individuals with known psychiatric illness
- Use cautiously in patients with a history of seizures or other factors that can lower the seizure threshold
- Discontinuing smoking may lead to pharmacokinetic or pharmacodynamic changes in other drugs the patient is taking, which could potentially require dose adjustment

Do Not Use

- If there is a proven allergy to varenicline

Special Populations

Renal Impairment

- Severe impairment: initial dose 0.5 mg once daily; maximum recommended dose 0.5 mg twice daily
- End-stage renal disease undergoing hemodialysis: maximum recommended dose is 0.5 mg once daily if tolerated
- Removed by hemodialysis

Hepatic Impairment

- Dose adjustment not generally necessary

Cardiac Impairment

- Effective in patients with cardiovascular disease; small increased risk of certain cardiovascular adverse effects in these patients

Elderly

- Some patients may tolerate lower doses better



Children and Adolescents

- Not shown to be effective for patients 16 years and younger



Pregnancy

- Effective June 30, 2015, the US FDA requires changes to the content and format of pregnancy and lactation information in prescription drug labels, including the elimination of the pregnancy letter categories; the Pregnancy and Lactation Labeling Rule (PLLR or final rule) applies only to prescription drugs and will be phased in gradually for drugs approved on or after June 30, 2001
- Controlled studies have not been conducted in pregnant women
- Pregnant women wishing to stop smoking may consider behavioral therapy before pharmacotherapy
- Not generally recommended for use during pregnancy, especially during first trimester

Breast Feeding

- Unknown if varenicline is secreted in human breast milk, but all psychotropics are assumed to be secreted in breast milk
- Recommended either to discontinue drug or bottle feed

The Art Of Psychopharmacology

Potential Advantages

- More effective than other pharmacotherapies for smoking cessation

Potential Disadvantages

- Not well studied in patients with comorbid psychiatric disorders

Primary Target Symptoms

- Cravings associated with nicotine withdrawal

Pearls

- More effective than nicotine or bupropion
- Unlike nicotine or bupropion, the patient cannot “smoke over” varenicline since varenicline, but not the others, will block the effects of additional smoked nicotine if the patient decides to smoke during treatment
- Although tested in the general population excluding psychiatric patients, where smoking rates in the USA are about 20–25%, there is great unmet need for smoking cessation treatments in patients

with psychiatric disorders, especially attention deficit hyperactivity disorder and schizophrenia, which have smoking rates in the USA as high as 50–75%

- Smoking cessation treatment in patients with comorbid psychiatric disorders is not well studied
- Preliminary results suggest that varenicline is not associated with worsening of psychiatric symptoms in stable patients with schizophrenia, schizoaffective disorder, or depression
- Patients with comorbid psychiatric disorders should be closely monitored in terms of their psychiatric symptoms, especially suicidality
- In a large multinational study requested by the FDA, there was no significant increase in moderate-to-severe neuropsychiatric adverse events with varenicline relative to nicotine patch or placebo in participants with or without history of psychiatric disorder

Suggested Reading

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Venlafaxine

Therapeutics

Brands

- Effexor
- Effexor XR
- *see index for additional brand names*

Generic?

Yes



Class

- Neuroscience-based Nomenclature: serotonin and norepinephrine reuptake inhibitor (SN-RI)
- SNRI (dual serotonin and norepinephrine reuptake inhibitor); often classified as an antidepressant, but it is not just an antidepressant

Commonly Prescribed for

(bold for FDA approved)

- **Depression**
- **Generalized anxiety disorder (GAD)**
- **Social anxiety disorder (social phobia)**

- **Panic disorder**
- Posttraumatic stress disorder (PTSD)
- Premenstrual dysphoric disorder (PMDD)

How the Drug Works

- Boosts neurotransmitters serotonin, norepinephrine/noradrenaline, and dopamine
- Blocks serotonin reuptake pump (serotonin transporter), presumably increasing serotonergic neurotransmission
- Blocks norepinephrine reuptake pump (norepinephrine transporter), presumably increasing noradrenergic neurotransmission
- Presumably desensitizes both serotonin 1A receptors and beta adrenergic receptors
- Since dopamine is inactivated by norepinephrine reuptake in frontal cortex, which largely lacks dopamine transporters, venlafaxine can increase dopamine neurotransmission in this part of the brain
- Weakly blocks dopamine reuptake pump (dopamine transporter), and may increase dopamine neurotransmission

How Long Until It Works

- Onset of therapeutic actions usually not immediate, but often delayed 2–4 weeks